AYMAN MAHDIA KHAN

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RESEARCH INTEREST

Hydrological Modeling, Flood Risk Management, Surface Water Management, Stormwater Management, Climate Change Adaptation, Groundwater Modelling

EDUCATION

B.Sc. in Civil Engineering, Chittagong University of Engineering & Technology (CUET) Feb 2020 – Jun 2025

- CGPA 3.21/4.00 (162.5 credits); merit position 36th of 130 (top 28%).
- Departmental **Merit Scholarship** for six semesters.

RESEARCH EXPERIENCE

Research Assistant, Dept. of Civil Engineering, CUET Nov 2025 – Present

"Integrated Assessment of Groundwater, Surface Water, and Rainwater Availability under Climate Change Scenarios in the Chattogram Hill Tracts (Integrated Water CHT)" | Project Leader: Prof. Dr. Aysha Akter | DRE-CUET funded (BDT 1.5M)

- Modelling water availability under RCP 4.5 and 8.5 across 13,295 km² of three hill districts, using **SWAT+ with BWDB rainfall, SRTM DEM and Sentinel-1/2 data**.
- Mapped surface-water (Rangamati) and groundwater (Bandarban) potential zones using frequency-ratio analysis in ArcGIS; two conference manuscripts submitted **(ICACE 2026, ICERIE 2027)**.
- **Led 14 field visits**: surveyed 33 springs, wells, and gauges; ran 26 household surveys, 3 focus groups, and 10 meetings with local and district government offices to turn model results into water-supply options.

Undergraduate Thesis (Grade: A+), Supervisor: Prof. Dr. Aysha Akter, CUET Apr 2024 – Jun 2025

"A Case Study on the Stormwater Drainage System Analysis and Design under Extreme Rainfall Events", Agrabad, Chittagong

- Built and calibrated an urban drainage model of the Agrabad catchment in **EPA SWMM 5.2** (94 sub-catchments, 245 conduits, 186 junctions) against observed flows and simulated a recent observed storm and a 50-year design storm.
- Redesigned **87 key conduits** that **reduced flooded junctions by 29.6%** and **peak flood volume by 39.9%** under both the recent and 50-year storm scenarios in Bangladesh's one of the most flood-exposed commercial districts.

PEER-REVIEWED JOURNAL PUBLICATION

- Akter, A., **Khan, A. M.**, & Farooq, S. M. (2026). *Surface water–groundwater–rainwater interactions in the Chittagong Hill Tracts, Bangladesh: A critical review of modeling and decision-support approaches*. *Hydrology, 13*(9), Article 242. https://doi.org/10.3390/hydrology13090242

CONFERENCE PAPERS

- Sultana, S., Shafique, S., Akter, S. T., Roy, C., **Khan, A. M.**, & Zishan, M. B. (2026, September 26). *Seismic risk assessment in Chattogram: An explainable machine learning approach*. iCONEECT 2026, Premier University, Chattogram, Bangladesh.
- Mahin, M. M. R., Hossain, H., **Khan, A. M.**, Alvi, M. M. I., & Banu, M. M. (2025, December). *A sustainable approach to develop eco-concrete using fly ash and silica fumes as SCMs*. ICERIE 2025, RUET, Rajshahi, Bangladesh.
- Srabony, S. C., Khan, M. S., **Khan, A. M.**, & Hameem, M. M. (2024, December). *Influence of industrial wastes and nano materials on the properties of recycled aggregate concrete: A review*. ICACE 2024, CUET, Chattogram, Bangladesh.

TECHNICAL SKILLS

Hydrological & hydraulic modelling: EPA SWMM 5.2, SWAT+ (research); MIKE 21, HEC-HMS, SewerGEMS (training)
GIS & remote sensing: ArcGIS (ArcMap 10.8), QGIS; SRTM DEM, Landsat, Sentinel-1/2

Programming & data analysis: Python (Pandas, GeoPandas, scikit-learn), machine learning (Random Forest, Gradient Boosting, XGBoost), Google Colab, MS Excel

Project & sustainability tools: MS Project, MS Visio, OpenLCA, Life-365

PROFESSIONAL TRAINING & FIELD EXPERIENCE

Industrial Training (credit-bearing, Grade A) | Institute of Water Modelling (IWM), Dhaka (3 weeks) Dec 2024

- Built 2D river morphology models in MIKE 21 (DHI), with data preparation and mapping in ArcMap 10.8.
- Trained in urban flood forecasting and water resources management in IWM's technical division.
- Visited the Goran Chatbari stormwater pumping station and Dasherkandi Sewage Treatment Plant.

Software-Aided Civil Engineering Design & Analysis | EDGE Project, BCC, ICT Division (80 hours) Jun 2024 – Oct 2024

- Government-funded training in GIS, HEC-HMS, SWMM, SewerGEMS, PLAXIS, ETABS and PMS.

Short courses & field visits: Hydrographic survey with Chittagong Port Authority (Feb 2025, 1 day); Sanitation for Urban Low-Income Communities, CUET & ITN-BUET, including a visit to Halishahar STP (Aug 2025, 2 days).

COMMUNITY ENGAGEMENT & SCIENCE COMMUNICATION

Flood Rehabilitation Project | ACI Student Chapter, CUET Oct 2024 – Jan 2025

- Led the construction of a home for a flood-displaced family in Sundarpur, Fatikchhari: managed site assessment, estimation, and costing on a BDT 100,000 budget, and coordinated approvals with the administration and local authorities.

Exhibitor, Bangladesh Innovation Fair 2026 | University Zone, CUET Civil Engineering Sep 2026

- Presented the Integrated Water CHT project with an interactive demonstration of groundwater, surface-water, and rainwater potential mapping, engaging over 700 visitors across three days.

AWARDS & COMPETITIONS

Top 20 international finalist (11th overall; 2nd from South Asia), American Concrete Institute (ACI) Spring 2022 Concrete Solutions Competition.

- Designed a flood-resistant modular house for flood-prone areas using Autoclaved Aerated Concrete (AAC) and Rice Husk Ash (RHA); the design first won the Intra-CUET Concrete Solutions Competition (ACI Student Chapter, 2022).

Champion, National Case Study Competition, Civil Explosion CUET, 2023.

- Designed an academic building for a forested site with minimal environmental impact.

Finalist, Face the Case 2.0, IEEE CUET Student Branch, 2022.

- Solved a real-world case addressing corruption and black ticketing in the railway system.

LEADERSHIP & SERVICE

General Secretary (May 2024 – May 2025) and Assistant General Secretary (Mar 2023 – May 2024), ACI Student Chapter, CUET; led a chapter of over 300 members; organised technical events, competitions, and the flood rehabilitation project.

Joint Organising Secretary, Green for Peace (Jul 2024 – Jun 2025), environmental advocacy, community outreach.

Judge, Envision 2.0, ASCE Student Chapter, CUET (Dec 2025); **Speaker**, Case Study Session, ASCE Student Chapter (Oct 2024); **Volunteer**, 7th ICACE (Dec 2024).

LANGUAGE PROFICIENCY

IELTS Academic (May 2026): Overall 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0); **Bangla (Native)**

REFERENCES

Dr. Aysha Akter
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Dr. G. M. Sadiqul Islam
Professor, Dept. of Civil Engineering, CUET
Email: gmsislam@cuet.ac.bd